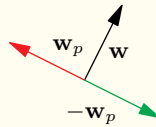


1.1.1

Gitt en vektor $\mathbf{w}$. Vi definerer $\mathbf{w}_p$ som vektoren som

- står vinkelrett på $\mathbf{w}$.
- peker til venstre med hensyn til retningen til $\mathbf{w}$.
- har lik lengde som $\mathbf{w}$.

Hvis $\mathbf{w} = [x, y]$, kan det vises at $\mathbf{w}_p = [-y, x]$, og at $-\mathbf{w}_p$ peker til høyre for $\mathbf{w}$.



Gitt vektorene $\mathbf{u} = [a, b]$, $\mathbf{u}_p = [-b, a]$, $\mathbf{v} = [c, d]$ og $\mathbf{v}_p = [-d, c]$.
Vis at

a) $\mathbf{u} \cdot \mathbf{v}_p + \mathbf{u}_p \cdot \mathbf{v} = 0$

b) $(\mathbf{u}_p)_p = -\mathbf{u}$

c) $(\mathbf{u} + \mathbf{v})_p = \mathbf{u}_p + \mathbf{v}_p$

Svar

a) $[a, b] \cdot [-d, c] + [-b, a] \cdot [c, d] = -ad + bc - bd + ad = 0$

b) $(\mathbf{u}_p)_p = ([-b, a])_p = [-a, -b] = -\mathbf{u}$.

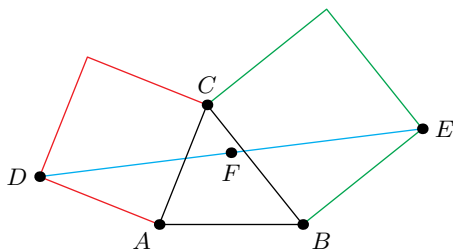
c)

$$\begin{aligned}(\mathbf{u} + \mathbf{v})_p &= ([a + c, b + d])_p \\&= [-(b + d), a + c] \\&= [-b, a] + [-d, c] \\&= \mathbf{u}_p + \mathbf{v}_p\end{aligned}$$

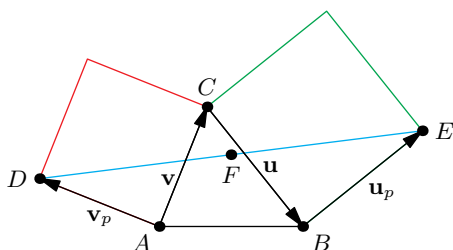
1.1.2

Den røde og den grønne firkanten er begge et kvadrat. Vi setter $\mathbf{v} = \overrightarrow{AC}$ og $\mathbf{u} = \overrightarrow{CB}$. F er midtpunktet på DE .

- Finn koordinatene til F uttrykt ved $\mathbf{u}$, $\mathbf{v}$, $\mathbf{u}_p$ og $\mathbf{v}_p$.
- Beskriv den eventuelle forflytningen til F hvis C forskyves i en vilkårlig retning. (Det holder å se på tilfellet hvor C ligger over AB).



Svar



Vi innfører et koordinatsystem med origo i A . Da er $D = \mathbf{v}_p$. Videre er $E = \mathbf{u} + \mathbf{v} + \mathbf{u}_p$. Videre er $\overrightarrow{DE} = -\mathbf{v}_p + \mathbf{v} + \mathbf{u} + \mathbf{u}_p$. Dermed er

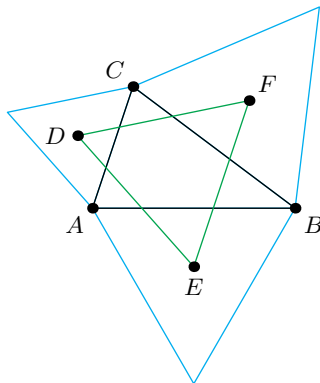
$$F = D + \frac{1}{2}\overrightarrow{DE} = \frac{1}{2}(\mathbf{v}_p + \mathbf{v} + \mathbf{u} + \mathbf{u}_p) = \frac{1}{2}(\mathbf{u} + \mathbf{v} + \mathbf{u}_p + \mathbf{v}_p)$$

$\mathbf{u} + \mathbf{v}$ og $\mathbf{u}_p + \mathbf{v}_p$ er ortogonale og har lik lengde. Så lenge $\mathbf{u} + \mathbf{v}$ ikke endres, vil altså ikke F forflyttes. En forskyvning av C medfører ingen endring av $\mathbf{u} + \mathbf{v}$, og dermed er posisjonen til F uavhengig av posisjonen til C .

1.1.3

De blå linjestykkene inngår i likesidede trekanter på utsiden av $\triangle ABC$. D , E og F er de respektive midtpunktene til de likesidede trekantene.

Vis at $\triangle DEF$ er likesidet.



Svar

Vi setter $\overrightarrow{AC} = 2\mathbf{v}$ og $\overrightarrow{AB} = 2\mathbf{u}$. Da er $\overrightarrow{BC} = 2\mathbf{v} - 2\mathbf{u}$. Videre har vi at

$$D = \mathbf{v} + k\mathbf{v}_p$$

$$E = \mathbf{u} - k\mathbf{u}_p$$

$$F = \mathbf{u} + \mathbf{v} - k(\mathbf{v}_p - \mathbf{u}_p)$$

hvor $k = \frac{1}{\sqrt{3}}$. Altså er

$$\overrightarrow{DF} = \mathbf{u} + k(\mathbf{u}_p - 2\mathbf{v}_p)$$

$$\overrightarrow{DE} = \mathbf{u} - \mathbf{v} - k(\mathbf{u}_p + \mathbf{v}_p)$$

$$\overrightarrow{EF} = \mathbf{v} + k(2\mathbf{u}_p - \mathbf{v}_p)$$

Da er

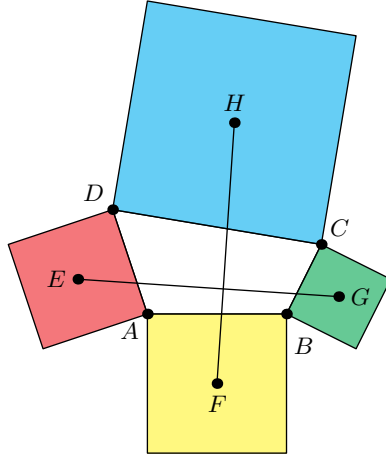
$$\begin{aligned} DF^2 &= \mathbf{u}^2 + 2k\mathbf{u}(\mathbf{u}_p - 2\mathbf{v}_p) + k^2(\mathbf{u}_p - 2\mathbf{v}_p)^2 \\ &= \mathbf{u}^2 - 4k\mathbf{u} \cdot \mathbf{v}_p + k^2(\mathbf{u}_p^2 - 4\mathbf{u}_p \cdot \mathbf{v}_p + 4\mathbf{v}_p^2) \\ &= \frac{4}{3}(\mathbf{u}^2 + \mathbf{v}^2 - \mathbf{u}_p \cdot \mathbf{v}_p) - 4k\mathbf{u} \cdot \mathbf{v}_p \end{aligned}$$

$$\begin{aligned}
DE^2 &= (\mathbf{u} - \mathbf{v})^2 - 2k(\mathbf{u} - \mathbf{v})(\mathbf{u}_p + \mathbf{v}_p) + k^2(\mathbf{u}_p + \mathbf{v}_p)^2 \\
&= \mathbf{u}^2 - 2\mathbf{u} \cdot \mathbf{v} + \mathbf{v}^2 - 2k(\mathbf{u} \cdot \mathbf{v}_p - \mathbf{u}_p \cdot \mathbf{v}) + k^2(\mathbf{u}_p^2 + 2\mathbf{u}_p \cdot \mathbf{v}_p + \mathbf{v}_p^2) \\
&= \frac{4}{3}(\mathbf{u}^2 + \mathbf{v}^2 - \mathbf{u}_p \cdot \mathbf{v}_p) - 4k\mathbf{u} \cdot \mathbf{v}_p
\end{aligned}$$

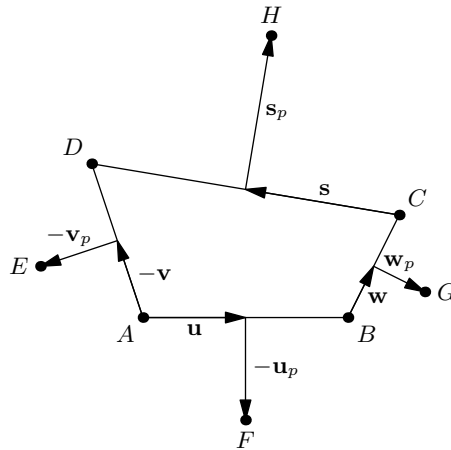
$$\begin{aligned}
EF^2 &= \mathbf{v}^2 + 2k\mathbf{v}(2\mathbf{u}_p - \mathbf{v}_p) + k^2(2\mathbf{u}_p - \mathbf{v}_p)^2 \\
&= \mathbf{v}^2 + 4k\mathbf{u}_p \cdot \mathbf{v} + k^2(4\mathbf{u}_p - 4\mathbf{u}_p\mathbf{v}_p + \mathbf{v}_p^2) \\
&= \frac{4}{3}(\mathbf{u}^2 + \mathbf{v}^2 - \mathbf{u}_p \cdot \mathbf{v}_p) - 4k\mathbf{u} \cdot \mathbf{v}_p
\end{aligned}$$

1.1.4

De fargede firkantene er kvadrater. E , F , G og H er de respektive midtpunktene til kvadratene. Vis at $EG = FH$ og $\overrightarrow{EG} \perp \overrightarrow{FH}$.



Svar



Vi innfører et koordinatsystem med origo i A , og setter $\overrightarrow{DA} = 2\mathbf{v}$, $\overrightarrow{AB} = 2\mathbf{u}$, og $\overrightarrow{BC} = 2\mathbf{w}$ (i figuren er $\mathbf{s} = \mathbf{u} + \mathbf{v} + \mathbf{w}$). Da er

$$E = -\mathbf{v} - \mathbf{v}_p$$

$$F = \mathbf{u} - \mathbf{u}_p$$

$$G = 2\mathbf{u} + \mathbf{w} - \mathbf{w}_p$$

$$H = \mathbf{u} + \mathbf{w} - \mathbf{v} + \mathbf{u}_p + \mathbf{v}_p + \mathbf{w}_p$$

Dermed er

$$\begin{aligned}\overrightarrow{EG} &= 2\mathbf{u} + \mathbf{w} - \mathbf{w}_p + \mathbf{v} + \mathbf{v}_p = 2\mathbf{u} + \mathbf{w} + \mathbf{v} + \mathbf{v}_p - \mathbf{w}_p \\ \overrightarrow{FH} &= \mathbf{w} - \mathbf{v} + 2\mathbf{u}_p + \mathbf{v}_p + \mathbf{w}_p\end{aligned}$$

Vi har at

$$\overrightarrow{FH}_p = \mathbf{w}_p - \mathbf{v}_p - 2\mathbf{u} - \mathbf{v} - \mathbf{w} = -\overrightarrow{EG}$$

Dette betyr at $\overrightarrow{FH}$ og $\overrightarrow{EG}$ er ortogonale. Videre er

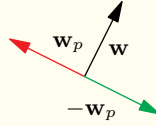
$$FH = |\overrightarrow{FH}_p| = |-\overrightarrow{EG}| = EG$$

1.1.5

Given a vector $\mathbf{w}$. We define $\mathbf{w}_p$ as the vector that

- is perpendicular to $\mathbf{w}$.
- points to the left with respect to the direction of $\mathbf{w}$.
- has the same length as $\mathbf{w}$.

If $\mathbf{w} = [x, y]$, it can be shown that $\mathbf{w}_p = [-y, x]$, and that $-\mathbf{w}_p$ points to the right of $\mathbf{w}$.



Given the vectors $\mathbf{u} = [a, b]$, $\mathbf{u}_p = [-b, a]$, $\mathbf{v} = [c, d]$, and $\mathbf{v}_p = [-d, c]$. Show that

a) $\mathbf{u} \cdot \mathbf{v}_p + \mathbf{u}_p \cdot \mathbf{v} = 0$

b) $(\mathbf{u}_p)_p = -\mathbf{u}$

c) $(\mathbf{u} + \mathbf{v})_p = \mathbf{u}_p + \mathbf{v}_p$

Answer

a) $[a, b] \cdot [-d, c] + [-b, a] \cdot [c, d] = -ad + bc - bd + ad = 0$

b) $(\mathbf{u}_p)_p = ([-b, a])_p = [-a, -b] = -\mathbf{u}$.

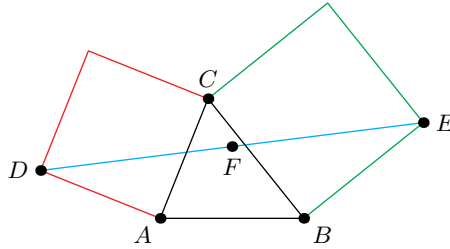
c)

$$\begin{aligned}
 (\mathbf{u} + \mathbf{v})_p &= ([a + c, b + d])_p \\
 &= [-(b + d), a + c] \\
 &= [-b, a] + [-d, c] \\
 &= \mathbf{u}_p + \mathbf{v}_p
 \end{aligned}$$

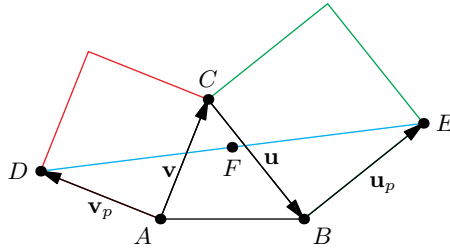
Problem: Bottema's theorem

The red and the green quadrilateral are both squares. F is the midpoint of DE . Let $\mathbf{v} = \overrightarrow{AC}$ and $\mathbf{u} = \overrightarrow{CB}$. Given a vector $\mathbf{w}$, we define $\mathbf{w}_p$ as the vector perpendicular to $\mathbf{w}$, pointing to the left with respect to the direction of $\mathbf{w}$, and with equal length as $\mathbf{w}$.

- Find the coordinates of F expressed in terms of $\mathbf{u}$, $\mathbf{v}$, $\mathbf{u}_p$, and $\mathbf{v}_p$.
- Describe the possible displacement of F if C is shifted in an arbitrary direction. (It is sufficient to consider the case where C is above AB).



Solution



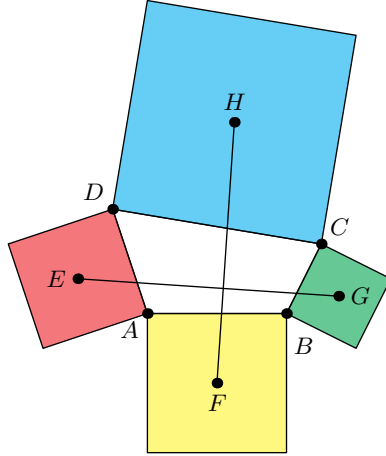
We introduce a coordinate system with the origin at A . Then $D = \mathbf{v}_p$. Furthermore, $E = \mathbf{u} + \mathbf{v} + \mathbf{u}_p$. Furthermore, $\overrightarrow{DE} = -\mathbf{v}_p + \mathbf{v} + \mathbf{u} + \mathbf{u}_p$. Thus,

$$F = D + \frac{1}{2}\overrightarrow{DE} = \frac{1}{2}(\mathbf{v}_p + \mathbf{v} + \mathbf{u} + \mathbf{u}_p) = \frac{1}{2}(\mathbf{u} + \mathbf{v} + \mathbf{u}_p + \mathbf{v}_p)$$

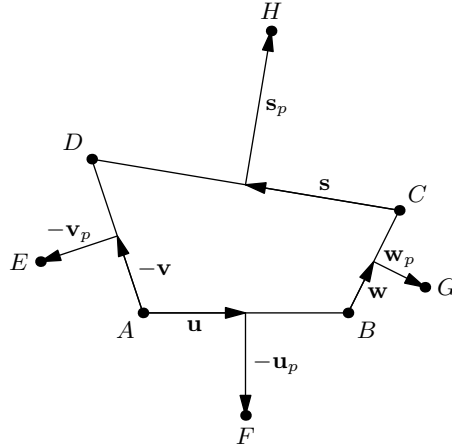
$\mathbf{u} + \mathbf{v}$ and $\mathbf{u}_p + \mathbf{v}_p$ are orthogonal and have the same length. As long as $\mathbf{u} + \mathbf{v}$ does not change, F will not be displaced. A displacement of C does not change $\mathbf{u} + \mathbf{v}$, and thus the position of F is independent of the position of C .

Problem: van Aubel's theorem

The colored quadrilaterals are squares. E , F , G , and H are the respective midpoints of the squares. Show that $EG = FH$ and $\overrightarrow{EG} \perp \overrightarrow{FH}$.



Answer



We introduce a coordinate system with the origin at A , and set $\overrightarrow{DA} = 2\mathbf{v}$, $\overrightarrow{AB} = 2\mathbf{u}$, and $\overrightarrow{BC} = 2\mathbf{w}$ (in the figure, $\mathbf{s} = \mathbf{u} + \mathbf{v} + \mathbf{w}$). Then,

$$\mathbf{E} = -\mathbf{v} - \mathbf{v}_p$$

$$\mathbf{F} = \mathbf{u} - \mathbf{u}_p$$

$$\mathbf{G} = 2\mathbf{u} + \mathbf{w} - \mathbf{w}_p$$

$$\mathbf{H} = \mathbf{u} + \mathbf{w} - \mathbf{v} + \mathbf{u}_p + \mathbf{v}_p + \mathbf{w}_p$$

Thus,

$$\begin{aligned}\overrightarrow{EG} &= 2\mathbf{u} + \mathbf{w} - \mathbf{w}_p + \mathbf{v} + \mathbf{v}_p = 2\mathbf{u} + \mathbf{w} + \mathbf{v} + \mathbf{v}_p - \mathbf{w}_p \\ \overrightarrow{FH} &= \mathbf{w} - \mathbf{v} + 2\mathbf{u}_p + \mathbf{v}_p + \mathbf{w}_p\end{aligned}$$

We have

$$\overrightarrow{FH}_p = \mathbf{w}_p - \mathbf{v}_p - 2\mathbf{u} - \mathbf{v} - \mathbf{w} = -\overrightarrow{EG}$$

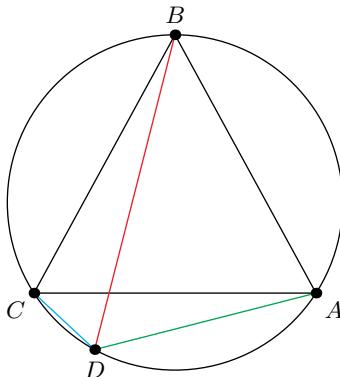
This means that $\overrightarrow{FH}$ and $\overrightarrow{EG}$ are orthogonal. Furthermore,

$$FH = |\overrightarrow{FH}_p| = |-\overrightarrow{EG}| = EG$$

A, B, C og D ligger på sirkelen, og $\angle ACB = \angle BAC = v$.

Vis at

$$BD = \frac{CD + DA}{2 \cos v}$$



Svar

$\angle BDC$ og $\angle ADB$ spanner over samme bue som henholdsvis $\angle ACB$ og $\angle BAC$. Alle disse er periferivinkler, og da $\angle ACB = \angle BAC$, har de alle like størrelse. Videre følger det at $BC = AB$. Av cosinussetingen har vi da at

$$CD^2 + BD^2 - 2CD \cdot BD \cos v = DA^2 + BD^2 - 2DA \cdot BD \cos v$$

$$CD^2 - DA^2 = 2BD \cos v (CD - DA)$$

$$(CD + DA)(CD - DA) = 2BD \cos v (CD - DA)$$

Antatt at $CD - DA \neq 0$, er

$$BD = \frac{CD + DA}{2 \cos v}$$

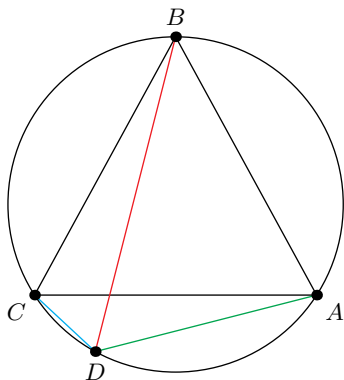
I tilfellet hvor $CD = DA$, er BD en diameter i sirkelen. Da har vi av Thales' setning omvendt at $\angle DCB = 90^\circ$. Altså er

$$BC = \frac{CD}{\cos v} = \frac{CD + DA}{2 \cos v}$$

A , B , C and D lies on the circle, and $\angle ACB = \angle BAC = v$.

Show that

$$BD = \frac{CD + DA}{2 \cos v}$$

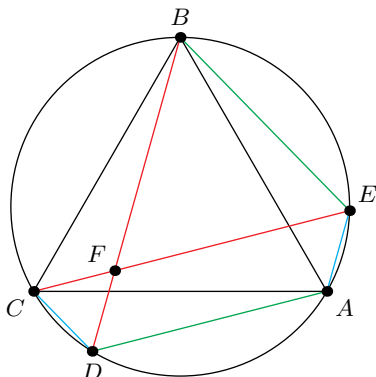


$\triangle ABC$ er likesidet, og $\triangle ACD \cong \triangle BAE$.

- a) Avgjør hvilke av disse vinklene som har lik verdi. Bestem vinkelverdien om det lar seg gjøre.

$$\angle BDC, \angle ADB, \angle CFD, \angle ACE, \angle BEC, \angle EFB \text{ og } \angle CAD$$

- b) Vis at $BD = CD + DA$.



Svar

- a) $\angle BDC = \angle BEC = \angle BAC = 60^\circ$ fordi de er periferivinkler som spenner over samme bue. Tilsvarende kan vi finne at $\angle ADB = 60^\circ$. $\angle CAD = \angle ACE$ fordi de er periferivinkler som spenner over sirkelsegmenter med lik lengde. Av vinkelsummen til $\angle ADC$ er $\angle DCA = 60 - \angle CAD$. Dermed er

$$\angle DCF = \angle DCA + \angle ACE = 60^\circ$$

Altså er $\triangle CDF$ likesidet. $\angle EFB = \angle CFD$ fordi de er topvinkler, og dermed er også $\triangle FEB$ likesidet. Følgelig har vi at

$$\angle BDC = \angle ADB = \angle CFD = \angle BEC = \angle EFB = 60^\circ$$

og

$$\angle CAD = \angle ACE$$

- b) Da $\triangle CDF$ og $\triangle FEB$ er likesidet, har vi at

$$BD = BF + FD = BE + CD = AD + CD$$